Literature Citation Verification System Based on DeepScientist

Project A — Engineering Track: Evidence Chain Module

Miaoyi Liao Xiaoyu Xiong Yuxiang Liu

STA304 Final Project · LLMs: From Principles to Applications

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The Problem

Core Pain Point

Research Agents (DeepScientist, OpenClaw) generate **professional-looking reports**, but users cannot answer three questions:

1. **Which conclusions come from real sources?** — citations untraceable
2. **Which are model inferences?** — no hallucination detection
3. **Which have no evidence at all?** — report wording stronger than evidence

Our Goal

Build an **auditable, traceable evidence-chain system** that verifies every claim:

Upload PDF → Structured Evidence ID → API Verification → G•Y•R• Scoring → Auto Audit

System Architecture

Pipeline Flow

Row 1 (Extraction)

`QQ User` ⟶ `crossdisc_idea (C)` ⟶ `parse_pdf (A)` ⟶ `verify_claim` $\times$N `(A)`

Row 2 (Scoring & Output)

`score_batch (B)` ⟶ `render_report (B)` ⟶ `artifact_record + memory_write (C)` ⟶ `QQ Returns RYG Report`

Three Roles

Person A — Tool Layer `claim_extractor`, `semantic_verifier` (PyPDF2, S2/arXiv/Crossref API), HARD STOP RULE in `builder.py`

Person B — Scoring Layer `traffic_light` (7-branch RYG), `render` (Markdown), MCP 5 tool registration, 6 Quest execution

FactCheck Pipeline Detail

Phase 0 — PDF Parsing

PyPDF2 extraction → regex detects "REFERENCES" heading → split by `[N]` / `[Author,Year]` → Unicode quote matching → truncation (200K chars, 40 claims max)

Phase 1 — Claim Verification

3-tier fallback: (1) Semantic Scholar, (2) arXiv Title API, (3) Crossref DOI search. Output: verdict + confidence + evidence snippet.

Phase 2 — Batch Scoring (B)

supported, conf $\geq$ 0.7	• Green
supported, conf $<$ 0.7	• Yellow
not_found, uncertain	• Yellow
contradicted, conf $\geq$ 0.7	• Red
contradicted, conf $<$ 0.7	• Yellow

Phase 3 — Render (B)

Rendered Markdown with: summary table (RYG counts), per-claim detail cards (verdict + confidence + evidence snippet), Evidence Chain table (E00X → C00X mapping).

Phase 4 — Persist (C)

`artifact__record` → evidence_store

`memory__write` → "episodes" kind.

Data Flow

Claim (C001) → VerificationResult → ScoredClaimResult → FactCheckResult

Enforcement — Dual-Layer

Layer 1 (SKILL.md): 5-State Machine

Evidence Chain & Audit

3 Evidence Entry Types

Type	ID	Metadata
QQ Text	E001	text + SHA256
QQ Image	E001-img	dimensions + SHA256
QQ PDF	E001-pdf	pages + SHA256 + full-text

Final Report Structure

- §1 FactCheck Results (rendered RYG table)
- §2 Evidence Chain Table (E00X ↔ C00X, 40 rows)
- §3–5 Cross-discipline bridges + hypothesis
- §6 Caveats (verifier limits, hedging

Audit: `audit_report()`

- Scans report for `[E001]`, `[E001-img]`, `[E001-pdf]` patterns
- Cross-references every cited ID against `evidence_store.json`
- Detects: fake IDs, unused evidence, bare claims (no annotation)
- Counts: supported / inferred / unverified claims
- Outputs PASS or WARN verdict

Three-Tier Annotation

`[E001]`	Evidence-supported claim (● green)
Inferred	Logically inferred, not directly proven
Needs Verification	Insufficient evidence, needs further check

Test Results: 6 Disciplines × 6 Quests

Quest	Discipline	Claims	verify	Score	Tools
027	ML/Federated	40	85	G17 Y18 R5	✓
028	Medical Edu.	8	27	G1 Y6 R1	✓
029	Chemistry	1	7	G1 Y0 R0	✓
030	Optimization	40	105	G10 Y28 R2	✓
031	Physics	40	94	G2 Y37 R1	✓
032	ML Survey	1	5	G0 Y1 R0	✓

Key Metrics

139 tests all passing | **996 events** | **323 verify_claim** calls | **36/36** mandatory tools

Failure Case Analysis

1. **Verifier False Positive (LIGO Q031)** C036 (GW190707) flagged ● contradicted. Cause: S2 matched different paper with overlapping authors. Handling: Caveats states "almost certainly a false negative."
2. **Abstract-Only Limitation (SGD Q030)** C018/C031 unverifiable — theorems need full-text access, abstracts lack detail.
3. **PDF Parser Fallback (Medical Q028)** `parse_pdf` returned 0 claims (PubMed-indexed, not arXiv). Agent auto-fallback to `bash_exec`. Extraction method marked as `bash_fallback` in evidence table.
4. **Agent Bypassed Toolchain (mid-dev, fixed)** Early runs: 10/40 verify_call, 0 score_batch, 0 render_report. Fix (3 iterations): MUST → 5-State Machine → HARD STOP RULE. Result: 36/36 compliance.

Boundary Cases

Case	Quest	Description
Minimal claims	Q029	Only 1 claim extracted; system correctly outputs PASS
Max uncertainty	Q031	37/40 ●; system honestly explains abstract-only limit
Cross-discipline refs	Q031	LIGO astro-ph format not recognized by S2; marked not_found
Internal proofs	Q027/030	Theorems/lemmas have no external citation target; correctly flagged
Multi-citation	Q027	`[20]--[24]` dash-range parsed into individual markers
Fake ID detection	test	`E999-img` caught by `audit_report` as non-existent

Before / After Comparison

Dimension	Before (Original)	After (Our System)
Message Recording	QQ logs, no structure	E00X evidence_store (16 fields)
Citation Annotation	None	[E001] + RYG tri-color labels
Reference Verification	None	S2 / arXiv / Crossref API
Image Evidence	No metadata	SHA256 + dims + format
PDF Evidence	Not supported	Pages + SHA256 + full-text sidecar
Audit Capability	Impossible	`audit_report` auto fake-ID check
Experiment Logging	None	`memory_write` structured schema
Test Coverage	0	**139 tests** / 100% pass
Grading Level	C+ (prompt only)	$\geq$ B- (MCP toolchain)

Limitations & Future Work

Current Limitations

- Abstract-only verification: most ● resolvable with full-text
- Verifier false positives: ∼5–10% ● may be misclassifications
- PyPDF2 weak on scanned PDFs
- API-dependent: offline mode not supported
- No video/audio support yet

Future Improvements

- Full-text verification via arXiv open-access
- Local model fallback (Ollama + sentence-transformers)
- Structured NLP pipeline for claim extraction
- Multi-connector support (WeChat, Discord)
- Caching layer to avoid duplicate API calls

Team Contributions

Member	Role	Core Contribution
Miaoyi Liao	Person A (Tool)	claim_extractor, semantic_verifier, HARD STOP RULE, evidence-chain store
Xiaoyu Xiong	Person B (Scoring)	traffic_light (7 branches), render (Markdown), MCP 5-tool registration, 6 Quest runs
Yuxiang Liu	Person C (Integration)	crossdisc_idea 5-Phase, QQ multi-modal evidence, daemon integration, memory schema

Cross-Collaboration Highlights

Phase 5 Enforcement: A (HARD STOP RULE) + B (5-State Machine) + C (memory/artifact schema) — three layers guarantee agent cannot bypass toolchain
Evidence Chain: A (storage + runner hooks) + B (audit + prompt protocol) + C (QQ multi-modal entry point) — full-link coverage

Summary

From "LLM Writes a Report" to "Agent Calls a Verifiable Toolchain"

Input	Verification	Output
6 disciplines	323 API lookups	RYG colored report
QQ real interaction	3-source fallback	Evidence Chain table
Text + Image + PDF	139 tests / 100%	Cross-disc. research ideas

Grade: $\geq$ **B-** (independent MCP toolchain, 36/36 mandatory calls)

`https://github.com/leoplasture/STA304_Final_Project`